Pamidronate

Pamidronate is used to treat and prevent problems caused by the effects of cancer on the bones.

On this page

- What is pamidronate?
- The effect of cancer on the bones
- Having pamidronate
- About side effects
- Side effects while treatment is being given
- Very common side effects
- Other side effects
- Other important information
- How we can help

What is pamidronate?

Pamidronate belongs to a group of drugs called bisphosphonates. It can be used to treat:

- high levels of calcium in the blood caused by cancer that has spread to the bones (secondary bone cancer)
- bone weakness or pain caused by myeloma or by breast cancer that has spread to the bones.

Pamidronate can be given with other cancer treatments.

More information about this treatment

This information is correct at time of publishing. But sometimes the types of cancer this treatment is used for, or treatment side effects, may change between revision dates.

You can talk to your cancer team if you want more detailed information about this treatment. Or visit the electronic Medicines Compendium (eMC) website, which has patient information leaflets (PIL) for individual drugs.

The effect of cancer on the bones

Bone cells constantly renew themselves. This helps bones keep their strength and shape.

There are 2 types of cell inside the bones. These are called:

© Macmillan Cancer Support. Reproduced with permission. Source: https://www.macmillan.org.uk/cancer-information-and-support/treatments-and-drugs/pamidronate-aredia · retrieved 26 October 2026

- osteoclasts, which break down and remove old bone
- osteoblasts, which build new bone.

As we get older, osteoclasts begin to break down more bone than osteoblasts make to replace it. This means our bones slowly become thinner (less dense). In some people, too much of the bone density is lost and this increases the risk of osteoporosis and bone fractures.

Some secondary bone cancers make a substance that causes osteoclasts to destroy more bone than can be replaced. The affected bone becomes weak and painful and can break more easily.

Bones contain calcium, which gives them strength. A bone affected by cancer may cause calcium from the bones to be released into the blood. A raised level of calcium in the blood is called hypercalcaemia. This may cause symptoms including:

- feeling sick (nausea)
- being sick (vomiting)
- feeling thirsty
- needing to pee (pass urine) more often
- constipation
- tiredness
- irritability
- confusion.

Having pamidronate

You have pamidronate as a drip into a vein (intravenously). You usually have it in a chemotherapy day unit or clinic as an outpatient. You may have it with other cancer drugs

During your course of treatment, you will meet with someone from your cancer team, such as a:

- cancer doctor
- chemotherapy nurse or specialist nurse
- specialist pharmacist.

This is who we mean when we mention doctor, nurse or pharmacist in this information.

During your course of treatment, you will have regular blood tests. This is to check that it is safe for you to have treatment. Your urine may also be checked.

Pamidronate may be given between 30 minutes to several hours depending on the dose. Your doctor or nurse will tell you how long your treatment will take. It is usually given every 3 to 4 weeks.

You may have pamidronate through 1 of the following:

- a cannula – a short, thin tube the nurse puts into a vein in the arm or hand
- a central line – a fine tube that goes under the skin of the chest and into a vein close by
-

© Macmillan Cancer Support. Reproduced with permission. Source: https://www.macmillan.org.uk/cancer-information-and-support/treatments-and-drugs/pamidronate-aredia · retrieved 26 October 2026

- a PICC line – a fine tube that goes into a vein in the arm and up into a vein in the chest
- an implantable port (portacath) – a disc that goes under the skin on the chest or arm and into a vein in the chest.

Your course of treatment

If you are having pamidronate to reduce pain or strengthen your bones, you may continue it for as long as it helps manage the symptoms.

If pamidronate is given to lower calcium levels, it is usually given as a single dose. Or you may be given several infusions over 2 to 4 days.

About side effects

We explain the most common side effects of this treatment here. We also include some that are less common.

You may get some of the side effects we mention, but you are unlikely to get all of them. And you may have some side effects, including rarer ones, that we have not listed here.

Other cancer treatments may cause different side effects. If you are also having other cancer treatment, you may have other side effects.

Always tell your doctor, nurse or pharmacist about any side effects you have. They can give you:

- drugs to help control some side effects
- advice about managing side effects.

It is important to take any drugs exactly as explained. This means they will be more likely to work for you.

Serious and life-threatening side effects

Some cancer treatments can cause serious side effects. Sometimes, these may be life-threatening. Your doctor, nurse or pharmacist can explain the risk of these side effects to you.

Contact the hospital

Your doctor, nurse or pharmacist will give you 24-hour contact numbers for the hospital. If you feel unwell or need advice, you can call at any time of the day or night. Save these numbers in your phone or keep them somewhere safe.

Dental check up before treatment

Rarely, this treatment can affect the jawbone. You can read more about this in the other side effects section below.

Having a healthy mouth and teeth reduces your risk of jaw problems. You usually need a dental check-up before you start treatment to check for any problems.

© Macmillan Cancer Support. Reproduced with permission. Source: https://www.macmillan.org.uk/cancer-information-and-support/treatments-and-drugs/pamidronate-aredia · retrieved 26 October 2026

If you need urgent treatment to lower high calcium levels in your blood, you do not usually need a dental check-up first.

Side effects while treatment is being given

Some people may have side effects while they are having a cancer drug or shortly after they have it:

Allergic reaction

Rarely, some people have an allergic reaction while having this treatment. Signs of a reaction can include:

- feeling hot or flushed
- shivering
- itching
- a skin rash
- feeling dizzy or sick
- a headache
- feeling breathless or wheezy
- swelling of your face or mouth
- pain in your back, tummy or chest.

Your nurse will check you for signs of a reaction during your treatment. If you feel unwell or have any of these signs, tell them straight away. If you do have a reaction, they can treat it quickly.

Sometimes a reaction happens a few hours after treatment. If you develop any of these signs or feel unwell after you get home, contact the hospital straight away on the 24-hour number.

Always call 999 if swelling happens suddenly or you are struggling to breathe.

Pain along the vein

This treatment can cause pain:

- at the place where the drip (infusion) is given
- along the vein.

If you feel pain, tell your nurse straight away. They can check the site. They may give the drug more slowly or flush it through with more fluid to reduce pain.

Very common side effects

These side effects happen to 10 or more people in every 100 people (10% or more) who have this treatment.

Flu-like symptoms

Some people have flu-like symptoms, such as:

- a mild fever (high temperature)
- chills
- pains in your muscles or joints.

Tell your doctor, nurse or pharmacist if you have these symptoms. They can give you advice.

Numbness or tingling

You may notice numbness or tingling around the mouth, or in the fingers and toes. Contact your doctor or the hospital straight away if you have these symptoms. This may be caused by low levels of calcium in your blood. You will have regular blood tests to check your calcium levels.

Your doctor may ask you to take calcium and vitamin D supplements, unless you are having this treatment to lower the levels of calcium in your blood. Your doctor will let you know if you need any supplements.

Muscle weakness

If you have low levels of phosphate in your blood, you may have muscle weakness. You may also feel confused. You will have regular blood tests to check your phosphate levels. Contact the hospital on the 24-hour number you have given if you have any of these symptoms.

Other side effects

These side effects happen to less than 10 in 100 people (less than 10%) who have this treatment. Some of them are much rarer than this but they are still important to know about. Rare means a side effect that happens to less than 1 in 1,000 people (less than 0.1%).

Low number of white blood cells

This treatment can reduce the number of white cells in your blood. These cells fight infection. If you have any signs of an infection, tell your doctor, nurse or pharmacist. These signs include:

- a cough
- a sore throat
- diarrhoea
- needing to pass urine (pee) often, or discomfort when you pass urine.

Contact the hospital straight away on the 24-hour number if you have any of these symptoms:

- a temperature above 37.5°C
- a temperature below 36°C
- you feel unwell, even with a normal temperature.

© Macmillan Cancer Support. Reproduced with permission. Source: https://www.macmillan.org.uk/cancer-information-and-support/treatments-and-drugs/pamidronate-aredia · retrieved 26 October 2026

Bruising and bleeding

This treatment can reduce the number of platelets in your blood. Platelets are cells that help the blood to clot.

If the number of platelets is low, you may bruise or bleed easily. You may have:

- nosebleeds
- bleeding gums
- heavy periods
- blood in your urine (pee) or stools (poo)
- tiny red, brown or purple spots that may look like a rash – these spots can be harder to see if you have black or brown skin.

If you have any unexplained bruising or bleeding, contact the hospital straight away on the 24-hour number. You may need a drip to give you extra platelets. This is called a platelet transfusion.

Anaemia (low number of red blood cells)

This treatment can reduce the number of red blood cells in your blood. Red blood cells carry oxygen around the body. If the number of red blood cells is low, this is called anaemia. You may feel:

- very low in energy
- breathless
- dizzy and light-headed.

If you have these symptoms, contact the hospital straight away on the 24-hour number. You may need treatment for anaemia. If you are very anaemic, you may need a drip to give you extra red blood cells. This is called a blood transfusion.

Increased bone pain

Sometimes pain in the affected bone can become worse for a short time when you start pamidronate. If this happens, your doctor can prescribe painkillers.

Feeling sick

You may feel sick during this treatment, but this is usually mild. Your doctor, nurse or pharmacist may prescribe you anti-sickness drugs to help prevent or control sickness. Take the drugs exactly as they tell you to. If you feel sick, take small sips of fluid often and eat small amounts regularly. It is important to drink enough fluids.

If you continue to feel sick, or if you are sick (vomit) 1 to 2 times in 24 hours, contact the hospital on the 24-hour number as soon as possible. They will give you advice.

Loss of appetite

This treatment can affect your appetite. Don’t worry if you do not eat much for 1 or 2 days. But if your appetite does not come back after a few days, or if you are losing weight, tell your doctor, nurse or pharmacist. They can give you advice. They may give you food or drink supplements. Or they may suggest changes to your diet or eating habits to help.

© Macmillan Cancer Support. Reproduced with permission. Source: https://www.macmillan.org.uk/cancer-information-and-support/treatments-and-drugs/pamidronate-aredia · retrieved 26 October 2026

Diarrhoea

This treatment may cause diarrhoea. Diarrhoea means passing more stools (poo) than is normal for you, or having watery or loose stools. You may also have stomach cramps. If you have a stoma, it may be more active than usual.

If you are passing loose stools 3 or more times a day and this is not normal for you, contact the hospital as soon as possible on the 24-hour number. Follow the advice they give you about:

- taking anti-diarrhoea medicines
- drinking enough fluids to keep you hydrated and to replace lost salts and minerals
- any changes to your diet that might help.

They might also ask you for a specimen of your stool to check for infection.

Constipation

This treatment can cause constipation. Constipation means that you are not able to pass stools (poo) as often as you normally do. It can become difficult or painful. Here are some tips that may help:

- Drink at least 2 litres (3½ pints) of fluids each day.
- Eat high-fibre foods, such as fruit, vegetables and wholemeal bread.
- Do regular gentle exercise, like going for short walks.

If you have constipation, contact the hospital on the 24-hour number for advice. They can give you drugs called laxatives to help.

If you have not been able to pass stools for over 2 days and are being sick, contact the 24-hour number straight away.

Tummy pain

You may get pain in your tummy (abdomen), or have indigestion. Your doctor, nurse or pharmacist can give you advice or treatment to help. Contact the hospital straight away on the 24-hour number if your symptoms do not go away, or get worse.

Effects on the eyes

Your eyes may become watery and feel sore. Your doctor, nurse or pharmacist can give you advice and eye drops to help with this.

Contact the hospital on the 24-hour number if:

- your eyes get red and inflamed (conjunctivitis)
- you have pain in your eyes
- you notice any change in your vision.

Skin changes

This treatment can cause a rash. Tell your doctor if this happens. They can give you advice and may prescribe creams or medicines to help. Any changes to your skin are usually temporary and improve

© Macmillan Cancer Support. Reproduced with permission. Source: https://www.macmillan.org.uk/cancer-information-and-support/treatments-and-drugs/pamidronate-aredia · retrieved 26 October 2026

when treatment finishes.

Blood pressure changes

This treatment can affect your blood pressure. It can cause high blood pressure and sometimes low blood pressure. Tell your doctor if you:

- have problems with your blood pressure
- have headaches
- feel dizzy.

Difficulty sleeping (insomnia)

Sometimes this treatment can affect your sleep. Tell your doctor, nurse or pharmacist if cancer treatment makes it difficult to sleep.

Headaches

This treatment may cause headaches. If you have headaches, tell your doctor, nurse or pharmacist. They can give you advice about painkillers that may help. Tell them if the headache does not get better, or gets worse.

Effects on the kidneys

This treatment can affect how the kidneys work. This is usually mild and goes back to normal after treatment ends. You will have blood tests to check how well your kidneys are working. Contact the hospital on the 24-hour number if you:

- have blood in your urine (pee)
- are passing less urine or peeing less often than usual.

Drinking fluids helps protect your kidneys. The advice is usually to try to drink at least 2 litres (3½ pints) of fluid each day. But follow any advice from your doctor, nurse or pharmacist about how much is right for you.

Jaw problems

Rarely, this treatment can affect the jawbone. Healthy bone in the jaw becomes damaged and dies. This is called osteonecrosis of the jaw (ONJ). It can cause:

- pain
- loosening of the teeth
- problems with the way the gums heal.

The risk of jaw problems is higher after some types of dental treatment. It is also higher in people who smoke, have gum disease or dentures that do not fit well.

It is important to avoid having dental treatment that could affect your jawbone when you are having bisphosphonates. This includes having a tooth or root removed or dental implants put in. You can still have fillings, gum treatments or a scale and polish.

© Macmillan Cancer Support. Reproduced with permission. Source: https://www.macmillan.org.uk/cancer-information-and-support/treatments-and-drugs/pamidronate-aredia · retrieved 26 October 2026

To reduce your risk of developing jaw problems, your doctor will advise you to:

- have a full dental check-up before starting treatment
- look after your teeth and gums during treatment (ask your dentist for advice)
- tell your dentist you are taking a bisphosphonate before having any dental treatment.

Tell your cancer specialist and dentist straight away if you at any time you develop:

- pain
- swelling or redness in your gums
- numbness or heaviness in your jaw
- loose teeth.

Pain in the thigh, hip or groin

Very rarely, people taking pamidronate develop a break (fracture) in their thigh bone without any obvious cause. Sometimes both thigh bones are affected.

If you have any thigh, hip or groin pain, tell your doctor and mention that you are taking bisphosphonates. They can arrange tests to check the thigh bones for any signs of weakness or fracture.

Muscle or joint pain

You may get pain in your muscles or joints for a few days after treatment. If this happens, tell your doctor, nurse or pharmacist. They can give you painkillers and advice. They can also tell you if any of the painkillers you usually take are suitable.

Tell them if the pain does not get better. Having warm baths and resting regularly may help.

Ear problems

Very rarely, bones in the outer ear may be affected by this treatment. Always tell your doctor if you have any:

- ear pain
- discharge from your ear
- ear infections.

You should also let them know if you notice any other changes in your ears or hearing.

Risk of having a fit (seizure)

You may have a higher risk of having a fit (seizure). But this is rare. This may be caused by high levels of calcium from the cancer and from having treatment. If this happens, call 999 straight away.

© Macmillan Cancer Support. Reproduced with permission. Source: https://www.macmillan.org.uk/cancer-information-and-support/treatments-and-drugs/pamidronate-aredia · retrieved 26 October 2026

Other important information

Going into hospital

If you are admitted to hospital for a reason not related to the cancer, it is important to tell the doctors and nurses that you are having treatment with bisphosphonates. You should tell them the name of your cancer specialist. This is so they can talk to them for advice.

Other medicines

Some medicines can affect how this treatment works or be harmful while you are having it. Always tell your cancer doctor, nurse or pharmacist about any drugs you are taking or planning to take, such as:

- medicines you have been prescribed
- medicines you buy in a shop, pharmacy or online
- vitamins or supplements
- herbal drugs and complementary or homeopathic therapies
- recreational drugs – for example, cannabis.

Pregnancy

Ask your doctor or pharmacist for advice before taking this medicine if you:

- are pregnant
- think you may be pregnant
- are planning to have a baby.

Breastfeeding

You are advised not to breastfeed while having this treatment, or for some time after treatment ends. This is because the drugs could be passed to the baby through breast milk.

Your doctor, nurse or pharmacist can give you more information.

Medical and dental treatment

If you need medical treatment for any reason other than cancer, always tell the healthcare professional that you are having cancer treatment. Give them the contact details for your cancer doctor or cancer team so they can ask for advice.

If you have appointments with a dentist, always tell them you are having cancer treatment. Talk to your cancer team before you have any dental treatment.

© Macmillan Cancer Support. Reproduced with permission. Source: https://www.macmillan.org.uk/cancer-information-and-support/treatments-and-drugs/pamidronate-aredia · retrieved 26 October 2026

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References

Visit the electronic Medicines Compendium (eMC) to download a Patient Information Leaflet (PIL) for more detailed information. The leaflet lists all known side effects.

Cancer Information Development team

Author

Date reviewed

Reviewed: 01 September 2026 | Next review: 01 September 2028

Our cancer information meets the PIF TICK quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about how we produce our information.

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

© Macmillan Cancer Support. Reproduced with permission. Source: https://www.macmillan.org.uk/cancer-information-and-support/treatments-and-drugs/pamidronate-aredia · retrieved 26 October 2026

You can read more about how we produce our information here.

How we can help

How to print a PDF from this website

Instructions on how you can make a PDF version of this page using the print function on your browser.

Clinical Information Nurse Specialists

Our Cancer Information Nurse Specialists are dedicated cancer nurses available to talk to on our Macmillan Cancer Support Line.

Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.

Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don’t want to call.

Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.

Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help